

Ch 6 → About algorithms manipulating data "hidden" in qubits.

MOTIVATION Deutsch's problem; $f: \{0,1\} \rightarrow \{0,1\}$ is either

constant, or balanced.
 $(f(0) = f(1))$ $(f(0) \neq f(1))$

To determine, how many times do we have to evaluate f ?

Classic sense : 2 times

Quantum sense : 1 times!

Review on preliminaries

Unit for information → Classic : Bit | Quantum : Qubit
 $|0\rangle, |1\rangle$ | $a|0\rangle + b|1\rangle$
 $a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1$

measurement → $|0\rangle$ with prob $|a|^2$
 $|1\rangle$ with prob $|b|^2$

Cannot be recovered → knowing a, b is impossible after measurement!

Space of qubit: $\mathbb{C}H$ (over $\mathbb{C}$)

Length n -qubit: $\mathbb{C}H^{\otimes n} := \mathbb{C}H \otimes \mathbb{C}H \otimes \dots \otimes \mathbb{C}H$ (n times)

e.g., $|010\rangle := |0\rangle \otimes |1\rangle \otimes |0\rangle \in \mathbb{C}H^{\otimes 3}$

Measurement rule : $X := \{|\phi_1\rangle, \dots, |\phi_n\rangle\}$: ONB of $\mathbb{C}H^{\otimes n}$, $|\psi_1\rangle, \dots, |\psi_m\rangle \in \mathbb{C}H^{\otimes m}$

$\sum_{k=1}^{2^n} \alpha_k |\phi_k\rangle |\psi_k\rangle \xrightarrow{\text{Measurement w.r.t. } X}$ Obtain $|\phi_k\rangle |\psi_k\rangle$ with prob $|\alpha_k|^2$.
 $(\alpha_k \in \mathbb{C}, \sum |\alpha_k|^2 = 1)$

* $|\psi\rangle$ is entangled if $|\psi\rangle$ is inseparable w.r.t. $\otimes$.

* $x \oplus y, x \odot y$: Binary addition & inner product.

* Regarding $\mathbb{C}H \hookrightarrow \mathbb{C}^2$,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

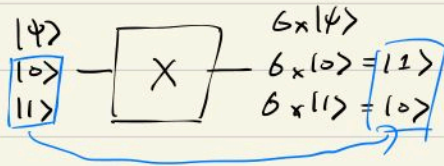
$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = \frac{\sigma_x + \sigma_z}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \left(\text{Rank} \begin{matrix} \sigma_x^2 = \sigma_z^2 = H^2 = id \end{matrix} \right)$$

(Bell states)

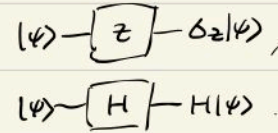
$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Def, Unitary operator $U: \mathbb{C}H^{\otimes n} \rightarrow \mathbb{C}H^{\otimes n}$ is called a **quantum n-gate**.

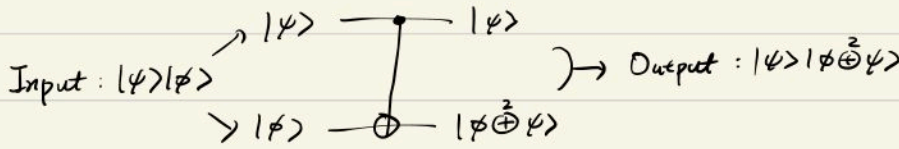
e.g., Quantum 1-gates



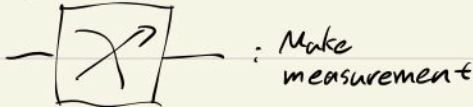
$X := \sigma_x$ is called a NOT-gate



e.g., Quantum 2-gate (C-Not gate)



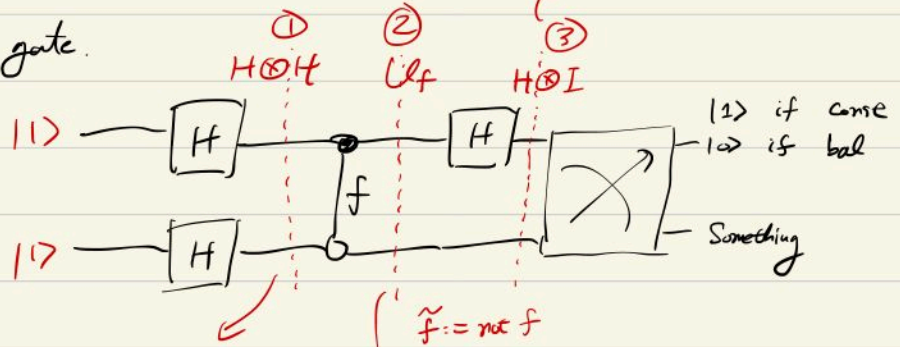
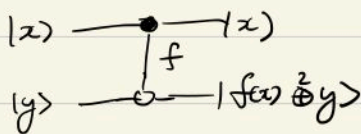
Other gates



x_0 : Activate $!$ iff $x_0 = 1$.

Solution of Deutsch's Problem:

Consider the f -CNOT (U_f) gate.



Rmk,

During ① $\rightarrow$ ②, f is evaluated simultaneously,

hence f might be regarded as it was evaluated once.

(So-called 'quantum sense')

$$\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(|00\rangle - |10\rangle - |01\rangle + |11\rangle)$$

Rmk, All computational BNB appears!

$$= \begin{cases} \frac{1}{\sqrt{2}}(|1\rangle(|f(0)\rangle - |\tilde{f}(0)\rangle)) & \text{if } f(0) = f(1) \\ \frac{1}{\sqrt{2}}(|0\rangle(|f(0)\rangle - |\tilde{f}(0)\rangle)) & \text{if } f(0) \neq f(1) \end{cases}$$

$$\frac{1}{2}(|0f(0)\rangle - |1f(0)\rangle - |0\tilde{f}(0)\rangle + |1\tilde{f}(0)\rangle)$$

$$= \begin{cases} \frac{1}{2}(|0\rangle - |1\rangle)(|f(0)\rangle - |\tilde{f}(0)\rangle) & \text{if } f(0) = f(1) \\ \frac{1}{2}(|0\rangle + |1\rangle)(|f(0)\rangle - |\tilde{f}(0)\rangle) & \text{if } f(0) \neq f(1) \end{cases}$$

* General Deutsch-Jozsa Problem

- Suppose $f: \{0,1\}^n \rightarrow \{0,1\}$ is either

constant, or balanced.

$$f(x) = \begin{cases} 0 & \text{for half of } x \in \{0,1\}^n \\ 1 & \text{for half of } x \in \{0,1\}^n \end{cases}$$

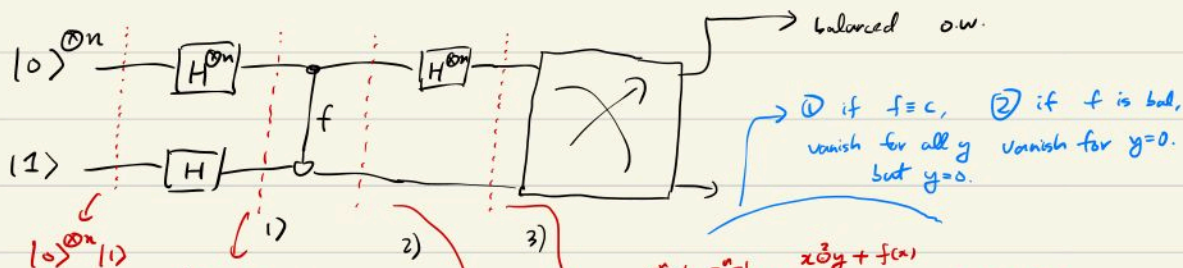
Q: Find the most efficient way to decide whether f is const or bal.

Algorithm

$$U_f: |x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle,$$

where $x \in \{0,1,\dots,2^n-1\}$, $y \in \{0,1\}$.

const if $|0\rangle^{\otimes n}$
balanced o.w.



$$\sum_{x=0}^{2^n-1} \frac{|x\rangle}{\sqrt{2^n}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Apply $H^{\otimes n}$

$$\sum_{x=0}^{2^n-1} \frac{(-1)^{f(x)} |x\rangle}{\sqrt{2^n}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$\sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} \frac{(-1)^{x \odot y + f(x)}}{2^n} |y\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= \begin{cases} (-1)^c |0\rangle^{\otimes n} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) & \text{if } f \text{ is const } f=c \\ \sum_{y=1}^{2^n-1} \alpha_y |y\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) & \text{if } f \text{ is bal.} \end{cases}$$

Rank: Every computational basis in $\mathbb{R}^{2^n}$ appears!

$$1) H^{\otimes n} |0\rangle^{\otimes n} = \sum_{x=0}^{2^n-1} \frac{|x\rangle}{\sqrt{2^n}} \quad (\text{See 5.83})$$

$$2) U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \frac{|x\rangle}{\sqrt{2}} (|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle)$$

$$= (-1)^{f(x)} \frac{|x\rangle}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Ex 6.67

$$3) \text{ When } x \in \{0,1\}, H|x\rangle = \sum_{y=0}^1 \frac{(-1)^{xy}}{\sqrt{2}} |y\rangle$$

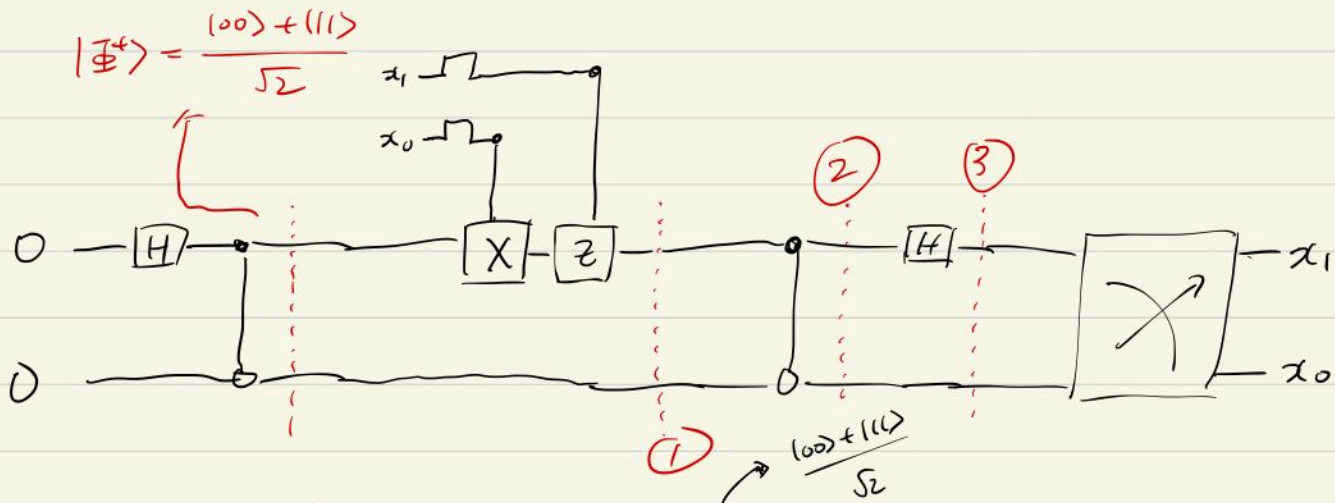
$$\Rightarrow \text{For } x \in \{0,1\}, H^{\otimes n} |x\rangle = \sum_{y=0}^{2^n-1} \frac{(-1)^{x \odot y}}{\sqrt{2^n}} |y\rangle$$

Sec 6.2 ~ 6.3 :

Interchange of information btw a qubit and 2-bit.

6.2. Dense Quantum Coding

- ① Alice has one of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, say $|x_1 x_0\rangle$.
- ② Alice do something and send a qubit
↳ depends on her bit
- ③ Bob measure Alice's qubit and
get a information corresponds to Alice's 2-bit.



If Alice's 2-bit is	we apply below to $ \Phi^+\rangle$	and at ① we have
00	$\text{id} \otimes \text{id}$	$ \Phi^+\rangle = 00 + 11$
01	$\sigma_x \otimes \text{id}$	$ \Phi^-\rangle = 01 + 10$
10	$\sigma_z \otimes \text{id}$	$ \Psi^+\rangle = 00 - 11$
11	$\sigma_z \sigma_x \otimes \text{id}$	$ \Psi^-\rangle = 01 - 10$

at ②

00 + 10

01 + 11

00 - 10

01 - 11

at ③ ($\sigma_x^2 = \text{id}$)

00

01

10

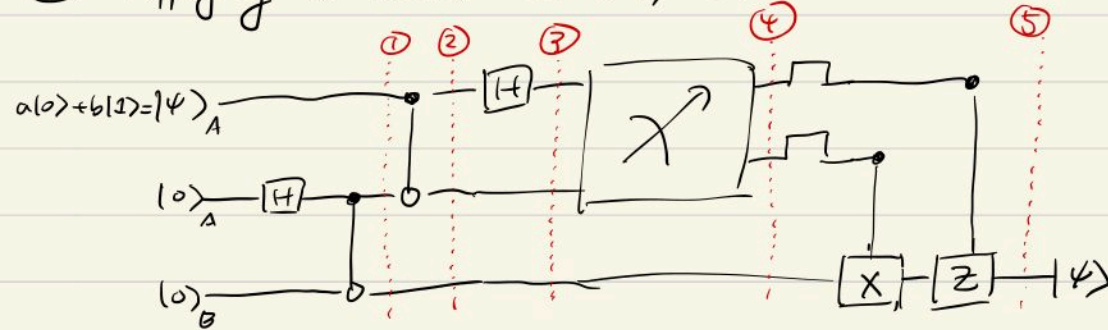
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6.3. Teleportation

* Copying a qubit is impossible, in principle. (Prop 4.11: No-Cloning Thm)

* However, it is possible to 'deliver' the information of a qubit via a 2-bit!

- ① Alice send a qubit $|\psi\rangle$ to some channel
- ② Alice measure the result and obtain some $|x_1 x_2\rangle|? \rangle$. ↗ Not deterministic
- ③ Alice tells Bob about $|x_1 x_2\rangle$ and Bob prepare an action, depending on $|x_1 x_2\rangle$.
- ④ Applying an action to $|? \rangle$, Bob has $|\psi\rangle$ on his register.



$$\textcircled{1} |\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = a|000\rangle + a|011\rangle + b|100\rangle + b|111\rangle$$

$$\textcircled{2} a|000\rangle + a|011\rangle + b|110\rangle + b|101\rangle$$

$$\textcircled{3} a(0+1)|00\rangle + a(0+1)|11\rangle + b(0-1)|10\rangle + b(0-1)|01\rangle$$

$$= 00(a0+b1) + 01(a1+b0) + 10(a0-b1) + 11(a1-b0)$$

$$= \frac{1}{2}(|00\rangle|\psi\rangle + |01\rangle\sigma_x^{-1}|\psi\rangle + |10\rangle\sigma_z^{-1}|\psi\rangle + |11\rangle\sigma_x^{-1}\sigma_z^{-1}|\psi\rangle) \dots \textcircled{4}$$

④ For equal probability, Alice has one of summand of $\textcircled{4}$, say $|x_1 x_2\rangle|? \rangle$.

⑤ Bob has $\sigma_z^{x_1} \sigma_x^{x_2} |? \rangle = |\psi\rangle$ for arbitrary x_1 and x_2 .

6.4. Quantum Cryptography

M : Set of messages, plaintext (we can read)

C : Set of encrypted messages (we can't read)

K : A set of (including unavailable) keys to access btw M, C .

Cipher: A protocol consist of encryption and decryption.
 $K \times M \rightarrow C$ $K \times C \rightarrow M$

* Quantum Key Distribution

- Way to generate a **random** bit sequence, which can be used as an encryption key.
- Based on the fact that qubit determine its post-measurement **randomly**.
- Protocols: BB84, Ek91 ...

Ex. Protocol BB84

Recall, For computational basis $|0\rangle, |1\rangle$ of $\mathbb{R}^H$,

i) $\{|\uparrow_z\rangle = |0\rangle, |\downarrow_z\rangle = |1\rangle\}$ (resp. $\{|\uparrow_x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, |\downarrow_x\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}\}$)

are e-vectors of observable δ_z (resp. δ_x) with e-values of 1 and -1.

ii) * Table of obtained e-value:

Qubit \ Observable	$ \uparrow_z\rangle$	$ \downarrow_z\rangle$	$ \uparrow_x\rangle$	$ \downarrow_x\rangle$
δ_z	1	-1	1 or -1	1 or -1
δ_x	1 or -1	1 or -1	1	-1

- After measurement, share observables and take bit seq. with some observables only.

For qubit no. 1	2	3	4	5	6	7	8	9	10	11	12	...
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Alice chooses randomly one of the observables σ_z or σ_x . Suppose she measures

Alice's observable σ_x σ_x σ_x σ_z σ_x σ_x σ_x σ_z σ_z σ_x σ_x σ_z ...

and observes

Alice's value +1 -1 -1 +1 +1 -1 +1 -1 +1 +1 +1 ...

The qubit is then in the

qubit state $|\uparrow_x\rangle$ $|\downarrow_x\rangle$ $|\downarrow_x\rangle$ $|\uparrow_z\rangle$ $|\uparrow_x\rangle$ $|\downarrow_x\rangle$ $|\uparrow_x\rangle$ $|\downarrow_z\rangle$ $|\uparrow_z\rangle$ $|\uparrow_x\rangle$ $|\uparrow_x\rangle$ $|\uparrow_z\rangle$...

Alice sends the qubits thus prepared to Bob. He chooses for each qubit randomly one of the observables σ_z or σ_x . Suppose he measures

Bob's observable σ_z σ_x σ_z σ_z σ_z σ_x σ_z σ_z σ_x σ_x σ_z σ_z ...

and observes

Bob's value -1 -1 -1 +1 +1 -1 +1 -1 +1 +1 -1 +1 ...

Alice and Bob publicly compare for which qubit they have measured which observable. But they do not reveal the outcome of the measurement, that is, the observed value. They thus divide the qubits into a set where they chose either the same observable or different ones. Measured

observables were $\neq$ $=$ $\neq$ $=$ $\neq$ $=$ $\neq$ $=$ $\neq$ $=$ $\neq$ $=$...

If they have chosen the same observable, their measured values have to agree. As a control they compare publicly every second of the observed values where they measured the same observable:

control-value Alice		+1		-1		+1	...
control-value Bob		+1		-1		+1	...

100% agreement in comparison of control values implies:

with a probability increasing with the number of control values the qubits have *not been read* between the measurements by Alice and Bob.

Use the observed values in the remaining cases where both measured the same observable as *joint*, *secret*, and *random*

bit sequence: -1 -1 +1 ...

- If α are same, then β must be same, but not: Eve detected!

33% disagreement in the control values implies eavesdropping.
Discard all qubits sent and start all over with a new sequence.

Sec 6.5. (Story only)

* RSA cipher: Security is based on the fact
it is hard to factorize large numbers.

* Uses two key: K_{pub} and K_{priv} .

* It suffices to know K_{pub} for encryption,
but decryption requires both K_{pub} and K_{priv} .

* For very large prime p, q , see $N = pq$.

* K_{pub} is known, and N can be extracted from K_{pub} .

* If we know p or q , we can extract K_{priv} .

* However, since factorization of large number is really hard
→ This guarantees the security!

- "Shor's algorithm" is a quantum algorithm
that allow factorization in short time.

(For detail, see sec 6.5.)

- Therefore, if Shor's algorithm can be
implemented, RSA will become insecure

- In fact, Shor's algorithm is a
special case of a wider class of
quantum algorithm, so-called a sol of
"Hidden subgroup problem".